RAMCO INSTITUTE OF TECHNOLOGY

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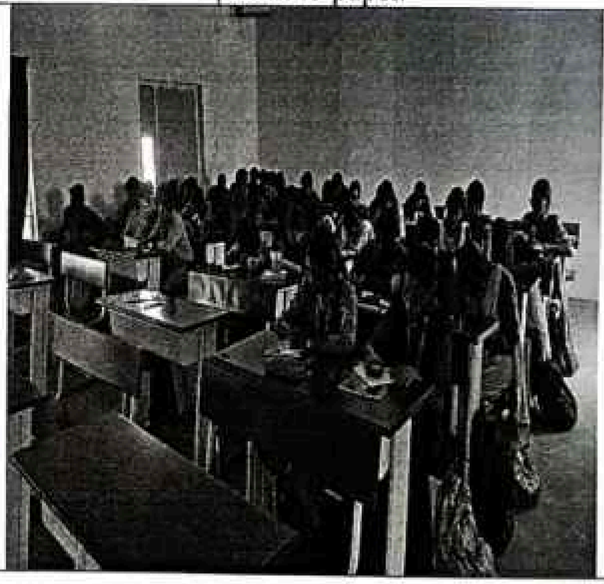
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (Odd Semester)

INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : VII Semester B.E. ECE B
Course Code & Title : EC8791 Embedded and Real Time Systems
Name of the Faculty member : Mr.A.Rameshbabu

Sl. No.	Date	Topic(s)	Activity*	Reference
UNIT I - Introduction to Embedded System Design				
1.	17.08.23	Designing with Computing Platforms	One Minute paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/



A.R
Signature of the Faculty Member

HOD

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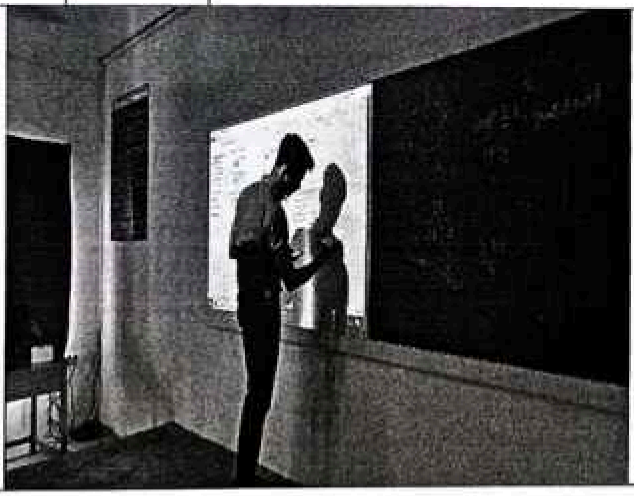
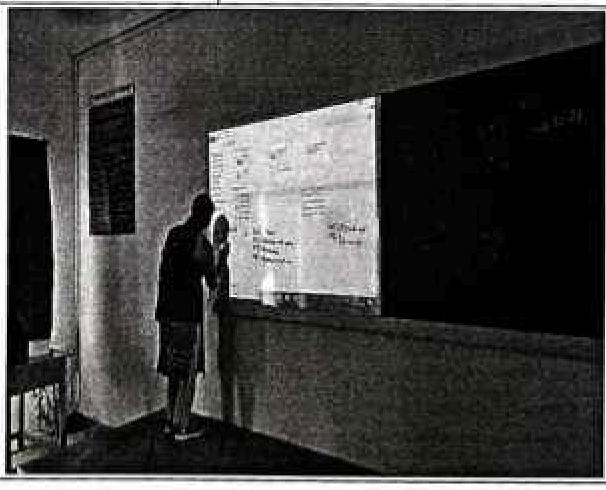
UNIT III - Embedded Programming				
1.	22.09.23	Models of Programs	Mind Map	https://www.ayoa.com/ourblog/6-mind-mapping-examples-for-students-and-teachers/





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Innovative Practice Description

- **Unit / Topic:** Unit II / Instruction Set
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 2.1
- **Activity Chosen:** Jigsaw
- **Justification:**
  - ➢ The basic concept of Design as a tradeoff can be explained with Jigsaw method. The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity.
  - ➢ The activity 'JIGSAW' is used to recall the concepts learnt and enhances the learning through this collaboration.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
  - ➢ **Materials for the Activity:**
    The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.
  - ➢ Plan for Implementing this Activity:
    - 5 groups are formed with 5 members
    - Each group is allotted with a name and it's known as Home group.
    - Each expert group is allotted with one problem.
    - The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas.
    - The Expert group members should go to their original shape group and discuss the points (30 Minutes) to the other members.

- All the members in home group learnt about all the topics through expert group members.
- Finally, one from each group should summarize the points of the topic Design as a tradeoff to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)

- **CO – PO / PSO mapping:**

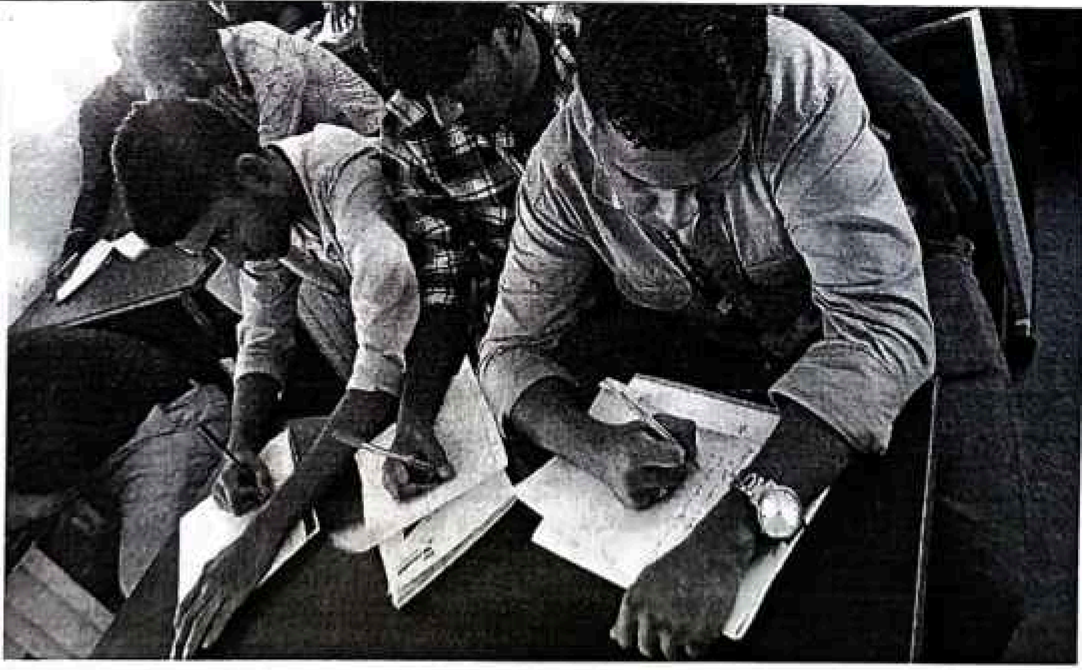
CO	PO 1	PO 2	PO 3	PO 9	PO 10	PO 12	PSO 1	PSO 3
CO3	3	3	3	1	1	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Jigsaw activity to discuss about various Instruction Sets, So Course outcome is mapped at level 1	In Jigsaw activity students will make effective explanations on various Instruction Sets, hence Course outcome is mapped at level 1	various Instruction Sets used to write code for ARM Processors to develop real time applications Embedded Systems, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students eagerly participated and enjoyed this activity.
  - ***Benefit of the practice:***
    - Students easily understand the concepts of various Instruction Sets.
    - Students can easily write their own code for any real life applications.
  - ***Challenges faced in implementation:***
    - Lack of preparation: Making the students to learn the materials is the challenging task.
    - Each student has to read the material posted in the course website.
    - Non participation in the activity: All the students should be made involve in the activity
    - Time management: The Jigsaw activity should be able to complete in the planned duration.

References:

- https://iitd-plos.github.io/col718/ref/arm-instructionset.pdf
- https://www.bravegnu.org/gnu-eprog/arm-iset.html
- http://www.ee.ncu.edu.tw/~jfli/computer/lecture/ch04.pdf

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Task Assignment
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 4.2
- **Activity Chosen:** Flipped Classroom
- **Justification:**
  - ➢ A flipped classroom is a type of blended learning where students are introduced to content at home and practice working through it at College.
- **Time Allotted for the Activity:** 50 Minutes

- **Details of the Implementation:**
  - ➢ This is the reverse of the more common practice of introducing new content at college, then assigning homework and projects to complete by the students independently at home.
  - ➢ In this blended learning approach, face-to-face interaction is mixed with independent study–usually via technology. In a common Flipped Classroom scenario, students might watch pre-recorded videos at home, then come to school to do the homework armed with questions and at least some background knowledge.
  - ➢ The concept behind the flipped classroom is to rethink when students have access to the resources they need most. If the problem is that students need help doing the work rather than being introduced to the new thinking behind the work, then the solution the flipped classroom takes is to reverse that pattern.

- **CO – PO / PSO mapping:**

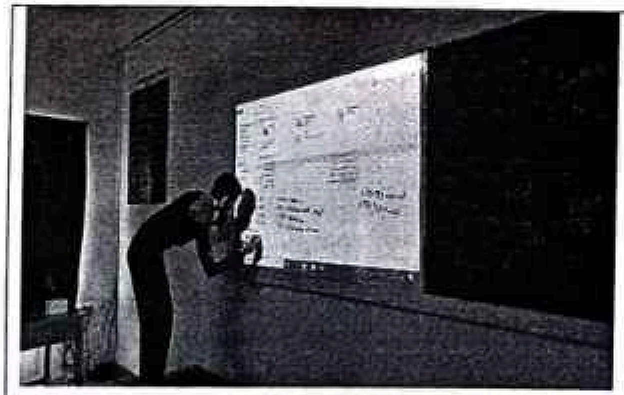
CO	PO 1	PO 2	PO 4	PO 9	PO 10	PO 12	PSO 1	PSO 2
CO3	3	2	2	1	1	1	1	3

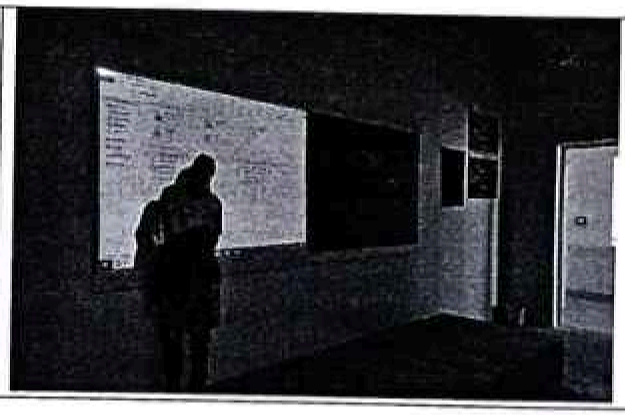
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Flipped Classroom activity to discuss about different Task Assignment Algorithms, So Course outcome is mapped at level 1	In Flipped Classroom activity students will make effective presentations about Task Assignment Algorithms, hence Course outcome is mapped at level 1	Knowledge of Task Assignment Algorithms is used to Design, analyze and develop any real time applications in the area of Embedded Applications, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ➢ Students eagerly participated and enjoyed this activity.
  - ❖ ***Benefit of the practice:***
    - ➢ Outcome attainment would have increased due to innovative practice over conventional practice.
    - ➢ Students easily understand the concepts of Task Assignment Process.
    - ➢ Students can answer easily about different types of Task Assignment Algorithms.

- ***Challenges faced in implementation:***
  - Faced issues while make the students to understand about Flipped Class Room Activity rules.
  - Make the students to learn about Task Assignment Algorithms before the start of this activity.

References:

- https://canvas.instructure.com/courses/5174605/files/folder/Notes/Unit%20IV
- the task set consists of periodic, preemptible tasks whose deadlines equal the task period.
- https://www.ibm.com/docs/en/warehouse-management/9.4.0?topic=management-task-assignment

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